Vinaykumar Venkateshkumar

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UK Student visa, coursework complete · Available immediately · Sponsorship may be required

Profile

Graduate design engineer for the engine externals and secondary-systems lane. Produces release-ready mechanical design from parametric CAD — detail and GA drawings with geometric tolerancing to ISO 1101, limits and fits to ISO 286, tolerance stack-up, sheet-metal flat patterns, DFM — and sizes it against load cases by FEA. Verifies geometry rather than assuming it, and treats a simulation result as a claim to be tested.

Technical Skills

Mech. Design	Geometric tolerancing (ISO 1101), limits and fits (ISO 286), tolerance stack-up (worst case and RSS), design for manufacture, sheet-metal forming (K-factor, bend allowance, flat patterns), detail and GA drawings
CAD	CATIA V5 (proficient, 80 hrs certified), CadQuery, pyOCC / Open CASCADE, ANSYS DesignModeler, STEP and DXF interchange, parametric geometry (iCST)
Analysis	FEA (CalculiX, gmsh, FreeCAD FEM): load cases, mesh convergence, stress-singularity identification, hand calculation as an independent check. CFD (ANSYS Fluent, ICEM, OpenFOAM): k-ω SST, grid independence. Python (NumPy, SciPy, pytest), MATLAB, Git
Domains	Gas turbine performance, turbomachinery aerodynamics, thermodynamics, aircraft structures, non-destructive testing

Selected Projects

Parametric Actuator and Accessory-Gearbox Generators, with Drawing Pack — *mechanical design, GD&T* *2026*

- Sized before drawn — a flight-control actuator family across four aircraft classes and an accessory gearbox across five power ratings (5–50 kW), with involute gear geometry, Lewis bending-stress sizing and SKF 60xx bearings — then issued as a four-sheet A4 pack: three details and a ballooned GA carrying ISO 286 fits (35 H8 bore, 21 f7 rod), ISO 1101 tolerances, a datum scheme and a five-contributor stack on 390 mm nominal (worst case ±0.90, RSS ±0.44).
- Verification found six real defects, including meshing teeth intersecting by 424 mm^3 in every exported assembly because the gears were never phased. The test suite was then checked against itself with a reintroduced sign error, which only the tooth-taper test catches. 132 automated tests.

Formed Sheet-Metal Equipment Bracket, Sized and Redesigned — *sheet metal, DFM, FEA* *2026*

- Flat pattern from a K-factor neutral axis rather than assumed (101.52 mm blank against 105.00 mm of summed legs), validated by conservation of volume to 0.25% against 5.12% for a naive blank. Five DFM rules, and a material trade with a real answer: 2024-T3 is twice as strong as 5052-H32 and cannot make the fold — 4T minimum bend radius against 1T.
- Sized against a 9g crash load in CalculiX, where a five-density convergence study made the result usable: peak stress climbed 12% and never settled — a singularity at a fixed constraint, so the 545 MPa headline is a boundary condition, not a stress. The converged 196.7 MPa against 193 MPa yield condemns the part marginally, not by 2.8×. Four fixes were screened analytically first — stiffening the upright makes that leg 7.4× stiffer, margin unchanged — before 2.0 mm gauge (126.1 MPa, +53%) and a re-issued drawing. 66 tests.

Nacelle Cowl Geometry and CFD Validation — *personal project, public data* *2026*

CST curve fitted to NASA TM 110300's published NACA 1-85-100 cowl ordinates to 0.019% RMS of max radius; a compressible RANS OpenFOAM case (k-ω SST, 108,904 cells) at Mach 0.79 matched to the report's wind-tunnel pressures to RMS C_p 0.65 over 17 stations, including where it misses the suction peak. *Separately: a Cranfield research project on nacelle installation aerodynamics, industrial context — result pending, confidential.*

Twin-Spool Cycle Model, and the NASA/GE E^3 Engine — *gas turbine performance* *2026*

Separate-exhaust station solver with every turbine pressure ratio matched to the power its own spool demands rather than assumed: 57.67 kN net thrust, 46.7% thermal efficiency, within 0.01% of the ideal-Brayton limit (82 tests). Separately, the E^3's published architecture, cycle, compressor blading and HPT flowpath transcribed with a page citation on every value, held by 37 tests forcing independent routes to agree.

Controlled Jets for Enhanced Mixing Rates — *BEng final-year project, team of 3* *Mar 2025*

Passive flow-control tabs in ANSYS Fluent across Mach 0.6–1.0, grid independence to 3.4M+ nodes, validated against experimental shadowgraph imaging: 64–76% core-length reduction against an uncontrolled baseline at Mach 0.8, no significant thrust penalty. *Further work at vinaykumar.is-a.dev* — bleed-air duct clearance study, airfoil panel method cross-checked against XFoil (0.047 RMS), parametric wing and blade-row generators.

Education

Postgraduate Coursework, Thermal Power and Propulsion — Cranfield University, Cranfield, UK *Oct 2025 – Sep 2026*
Gas Turbine Performance · Combustion · Turbomachinery Aerodynamics · Propulsion System Design

BEng Aeronautical Engineering — CGPA 7.37/10 (first class) — KCG College of Technology, Chennai, India *2021 – 2025*

Certifications: CATIA V5 (Proficient), CADD Centre, 80 hrs, 2024 · MATLAB Essential Training, LinkedIn Learning, 2024 · IELTS Academic 6.5, UKVI B2, Jan 2025

Experience

Engineering Intern, Aircraft Maintenance — AIESL — Boeing 737 MRO base, India *Mar – Apr 2025*
Rotation across Component Overhaul, Production Planning and Stores servicing Air India Express. Documented wheel/brake overhaul to specification (torque 158 lb-ft, tyre pressure 205 ± 5 psi) and Eddy Current Testing; tracked parts in AMOS and RAMCO.